Lesson notes · for you

Tables, keys and types

Data engineering · Level 3 · Lesson 1

Time
17 min

Materials
Your phone with Darsi, for the SQL steps; Paper and a pen, for sketching tables

Learning goals

- Explain why apps use a database.
- Design tables with primary keys, types and foreign keys.
- Write a CREATE TABLE and a first SELECT.

Activity: Split the wide table

1. Look (3 min): Look at a wide table of pupils, their class, the class teacher and her phone. Circle every fact that appears more than once.
2. Split (6 min): Redraw it as a pupils table and a classes table. Mark each primary key and the foreign key.
3. Change (4 min): Change the teacher's phone in both designs. How many cells did each need?
4. Write (4 min): Write the CREATE TABLE for your classes table, with a type for each column.

Use made-up names and numbers.

Discussion

- Why are names bad primary keys?
- What went wrong when a fact was stored twice?
- Which column in likes points to photos?

Check yourself

- Sketch two linked tables for an app, and mark the primary and foreign keys.

Try it this week
Pick an app you use, like a messaging app. Sketch its tables: what is a row in each, and what's the key? Which tables link?

Worksheet

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Name _____ Class _____ Date _____

1 **Why store the class teacher's name in a classes table, not in every pupil's row?**

◯ Because tables must be small ◯ So it's harder to find ◯ So a change happens in one place

2 **Put the lines in order to create the likes table, with its columns in this order: id, user_id, photo_id, liked_at. id is the primary key; the other two ids are foreign keys.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `);`

☐ `CREATE TABLE likes (`

☐ `id INTEGER PRIMARY KEY,`

☐ `user_id INTEGER REFERENCES users(id),`

☐ `user_name TEXT,`

☐ `liked_at TEXT`

☐ `photo_id INTEGER REFERENCES photos(id),`

3 **id is the primary key of users. At most how many rows can this query return?**

```
SELECT name FROM users WHERE id = 7;
```

◯ As many as share a name ◯ One ◯ Every row

4 **Show the captions of all the photos posted by user 3 (Chipo).**

```
SELECT caption FROM photos;
```

Write your program:

5 **Match each word to its meaning.**

Primary key	• • • •	Points to a row in another table
Foreign key	• • • •	INTEGER or TEXT, for each column
Schema	• • • •	Picks out one row of its own table
Type	• • • •	The design of all the tables and links

✎ **Pick an app you use every day. What would be a row in its main table, and what would its primary key be?**

Check your answers

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1 **So a change happens in one place**

Stored once, it's changed once, and every pupil's row stays right through the link to the classes table.

2 **Top to bottom: 6, 1, 2, 3, X, 5, 4**

```
CREATE TABLE likes (
    id INTEGER PRIMARY KEY,
    user_id INTEGER REFERENCES users(id),
    photo_id INTEGER REFERENCES photos(id),
    liked_at TEXT
);
```

Each like has its own id (the primary key), points to one user and one photo with foreign keys, and records when it happened.

3 **One**

A primary key is different in every row, so WHERE id = 7 matches one row at most (or none, if there's no user 7).

4

```
SELECT caption FROM photos
WHERE user_id = 3;
```

Each photo row holds the id of the user who posted it, so WHERE user_id = 3 keeps Chipo's three photos.

5

Primary key — Picks out one row of its own table
Foreign key — Points to a row in another table
Schema — The design of all the tables and links
Type — INTEGER or TEXT, for each column

Keys identify and link rows, types say what each column holds, and the schema is the whole design.